

Forklift Crane Attachment

Design and Analysis of an Adjustable, Demountable Lifting Boom



The crane attachment shown in operating context. The forklift vehicle is a reference model; the attachment is the project scope.

COURSE	Machine Design I
DESIGN	Adjustable crane attachment for a 3-ton forklift
AUTHORS	Ali Sakhaei and Mohammadreza Sotude
TOOLS	SolidWorks, analytical strength calculations, cost estimation

1. Design Problem and Requirements

The objective was to design a crane boom that mounts on the forks of a 3-ton forklift and extends its ability to lift loads that cannot be reached directly. The attachment had to remain manually adjustable, mechanically simple, transportable, and fully demountable.

Functional requirements

Requirement	Specified value or condition
Host vehicle	3-ton forklift
Fork geometry	1,000 mm long, 120 mm wide, 40 mm thick
Fork spacing	800 mm center-to-center during installation
Maximum hook reach	2,500 mm beyond the fork tips
Boom angle	0°, 15°, 30°, and 45°
Hook-position increments	200 mm along the adjustable extension
Stability criterion	Forklift stability evaluated with a factor of safety of 1.5
Assembly	Detachable components that can be transported and assembled on site

Design scope

- Conceptual design of the base frame, boom, telescoping extension, hook, pins, and fork connections.
- Static calculations for the forklift stability limit and critical load cases.
- Sizing of bolts, pins, welded joints, structural tubes, channels, angles, and plates.
- SolidWorks part and assembly models with manufacturing and assembly drawings.
- Bill of materials, assembly procedure, product sheet, and approximate fabrication cost.

The project concerns the crane attachment. A pre-existing forklift CAD model was used only to show the installation context and is not included among the published design files.

2. Load Cases and Structural Analysis

The allowable lifted load changes with hook reach and boom angle. The design therefore considers multiple boom positions rather than a single rated load. Forklift stability provides the first limit, followed by stress and joint checks for the attachment itself.

Stability and member checks

$$SF_{stability} = \frac{M_{resisting}}{M_{overturning}} \geq 1.5$$

$$\sigma_b = \frac{Mc}{I} \qquad \tau = \frac{VQ}{It}$$

The boom and support members are evaluated for bending, shear, local bearing, and section strength. The telescoping arrangement requires checks at both maximum extension and the discrete pin locations because these positions change the bending moment and the contact forces.

Connection checks

$$\tau_{bolt} = \frac{F}{nA} \qquad \sigma_{bearing} = \frac{F}{td}$$

$$SF = \frac{S_y}{\sigma_{equivalent}}$$

Connection	Elements evaluated	Primary failure modes
Fork-mounted base	M24 and M20 bolts, plates, channels	Bolt shear, bearing, plate bending
Boom pivot	M16 clevis pin and supporting plates	Double shear, bearing, bending
Extension lock	M10 clevis pin and liners	Pin shear, local bearing, tube deformation
Hook connection	Grade 80 hook and shackle	Rated capacity and pin loading
Welded subassemblies	Multiple fillet-weld sizes	Throat shear and combined loading

The calculations are retained in the original academic report. This portfolio edition summarizes the design logic and does not replace a certified lifting-device calculation or inspection.

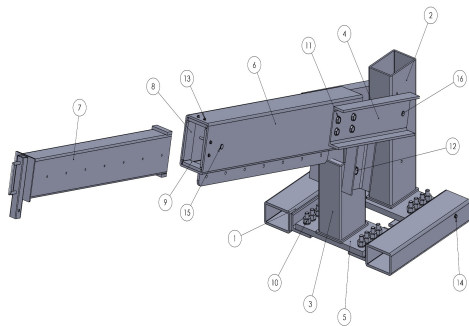
3. Structural Components and Materials

The structure combines hollow sections, channels, angles, and plates. Standard fasteners and removable pins allow the attachment to be assembled, adjusted, and disassembled without permanent connection to the forklift.

Component	Section or material	Function
Outer boom	260 × 180 × 10 mm rectangular hollow section	Primary bending member
Inner extension	200 × 120 × 8 mm rectangular hollow section	Adjustable hook reach
Base profile	200 × 120 × 12.5 mm rectangular hollow section	Transfers loads to fork-mounted frame
Channel	UAP 200, 2,100 mm long	Base-frame load distribution
High-strength plate	StE500, 570 × 530 × 30 mm	Critical connection region
General plates	St37, several thicknesses	Brackets, liners, and support plates
Angle	L 150 × 90 × 15 mm	Fork-frame support

Selected standard hardware

- Twenty-four M24 bolts for the main bolted structure, including specified strength classes.
- Eight M20 grade 8.8 bolts and eight M8 flange screws.
- M16 and M10 clevis pins with spring cotters.
- Grade 80 hook and screw-pin shackle.
- Fillet welds with several specified throat sizes for the welded subassemblies.



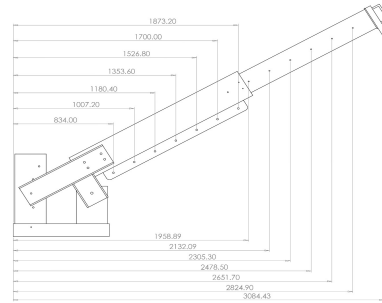
SolidWorks model of the crane attachment without the third-party forklift model.

4. CAD Model and Manufacturing Documentation

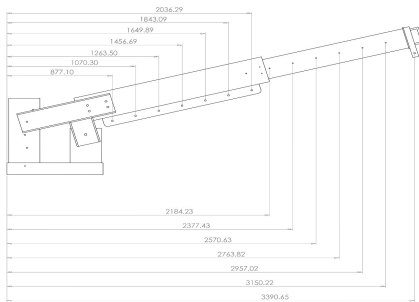
The SolidWorks model is divided into a fork-mounted base, pivoting boom, telescoping inner tube, liners, pins, plates, and connection hardware. Separate assembly and part files support editing and drawing generation.



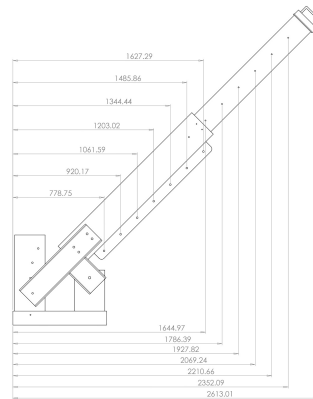
Rendered attachment



Boom-position drawing



Adjustment configuration



Extended and inclined configuration

Drawing package

The published files include SolidWorks parts and assemblies, native drawing files, and PDF exports for the main base and boom assemblies. Drawings cover the primary components, assembly relationships, discrete boom angles, and mass configurations.

The original folder relationships are preserved to reduce broken references when the assemblies are opened. The unrelated forklift reference assembly and its component files are excluded.

5. Assembly Strategy and Cost Estimate

The design separates the product into transportable subassemblies. The base frame is positioned over the forklift forks and secured through bolted connections. The outer boom is attached to the base through the main pivot. The telescoping member is inserted into the boom and locked at a selected reach with a removable pin.

Assembly sequence

- 1 Prepare the welded base-frame and boom subassemblies from the specified profiles and plates.
- 2 Position the base profiles over the two forklift forks and install the structural fasteners.
- 3 Connect the outer boom to the base using the pivot pin and cotter.
- 4 Insert the telescoping extension, liners, handle, and locking plates.
- 5 Select the required extension and boom angle, then secure the corresponding pins and fasteners.
- 6 Install the shackle and rated hook and perform a pre-use joint and alignment inspection.

Cost estimate

Item	Approximate value	Basis
Steel sections, plates, and hardware	26.82 million toman	Reported market prices in late May and early June 2025
Welding	0.34 million toman	Estimated weld lengths and size-dependent rates
Epoxy coating consumed	0.83 million toman	About 2.64 kg for a 100 μm dry film over 12 m^2
Rounded fabricated structure	Approximately 28 million toman	Academic estimate; excludes time-dependent purchasing variation

Prices are historical and location-dependent. They are included to demonstrate design-for-manufacture and cost awareness, not to provide a current quotation.

6. Conclusions and Repository Scope

The project develops a complete mechanical attachment from requirements to a manufacturable CAD package. Its strongest aspect is the combination of stability assessment, component and joint calculations, adjustable geometry, assembly planning, drawings, and cost estimation.

Main outcomes

- A demountable crane attachment compatible with the specified forklift fork geometry.
- Manual adjustment of hook reach up to 2.5 m and boom angle through four discrete positions.
- Analytical checks for structural members, bolts, pins, bearing regions, and welds.
- A detailed SolidWorks model with manufacturing and assembly drawings.
- A bill of materials, assembly sequence, and approximate fabrication cost.

Repository contents

Folder	Contents
solidworks/	Original attachment parts, assemblies, drawings, and PDF drawing exports
images/	Clean project renders and selected configuration drawings
docs/	English portfolio report

Limitations and safety

This is an academic design and has not been fabricated, proof-load tested, certified, or approved for lifting operations. A real implementation would require verified material certificates, code-based lifting-device calculations, weld-procedure qualification, fatigue and buckling checks, forklift-manufacturer approval, physical testing, inspection requirements, and review by a qualified engineer.

Authorship and external assets

The crane attachment was developed jointly by Ali Sakhaei and Mohammadreza Sotude. The vehicle model used in contextual renders was a pre-existing reference asset and is not included in the repository or presented as original work.